

Arpit Verma

📞 7052501218 | ✉ av6821246@gmail.com | 🌐 github.com/Arpit1825 | 🔗 linkedin.com/in/arpit-verma141 | 📁 portfolio-arpit-verma.vercel.app

CAREER OBJECTIVE

Published patent holder and Computer Science (AI & ML) undergraduate with hands-on experience in MERN stack development, data structures & algorithms, and machine learning fundamentals. Built and deployed 3 real-world projects using React.js, Node.js, and MongoDB Atlas. Passionate about software engineering, problem-solving, and building scalable applications.

SKILLS

Programming Languages: Python, C, C++

Frontend: HTML5, CSS3, JavaScript, React.js, Tailwind CSS

Backend: Node.js, Express.js, REST APIs

Databases: MySQL, Firebase, MongoDB, Mongoose

Libraries & Tools: Socket.io, JWT, bcrypt, Axios, Chart.js, NumPy, Pandas, Matplotlib, Redux Toolkit

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Postman

Core Concepts: Data Structures & Algorithms, OOPs, Operating Systems, DBMS, API Integration

PROJECTS

Smart Water Automation and Dosing System | (Sept 2025 – June 2026)

GitHub | Live Demo

- Developed a **patent-published IoT-based automation system** integrating **ESP32** and **Python** for real-time monitoring and control.
- Created a **MERN web-based dashboard** enabling real-time monitoring, control, and **data visualisation**.
- Executed automated decision-making logic for chemical dosing using real-time sensor inputs.
- Achieved **patent publication in Indian Patent Journal (Feb 2026)**.

Smart Queue Management System | (Jan 2026 – June 2026)

GitHub | Live Demo

- Developed and deployed a full-stack Smart Queue Management System using Node.js, Express.js, MongoDB Atlas, and **Socket.io**.
- Applied **JWT** authentication, **QR code**-based token tracking, and real-time queue status updates.
- Developed a multi-counter queue allocation system with live monitoring of waiting, serving, and completed tokens.
- Built an analytics dashboard and deployed the application on **Render** with **MongoDB Atlas** integration.

CryptoTracker: Real-time Cryptocurrency Tracker | (Feb 2026 – March 2026)

GitHub | Live Demo

- Developed a high-performance **React.js** web application to track **500+ cryptocurrencies** with real-time market data.
- Integrated real-time cryptocurrency market data APIs to display live prices, trends, and market statistics.
- Integrated **CoinGecko API** using **Axios** to fetch dynamic data, including price, market cap, and 24-hour fluctuations.
- Implemented **global state management** using **Redux Toolkit**, ensuring efficient data flow and minimising unnecessary re-renders.
- Developed a **responsive, mobile-first UI** using **Tailwind CSS**, enhancing accessibility across devices.
- Built **interactive data visualisations** using **Chart.js** for historical trend analysis (24h, 7d, 30d).

EDUCATION

Pranveer Singh Institute of Technology, Kanpur | 3rd Year

Bachelor of Technology in Computer Science (AI&ML).

(2024-2028)

Coursework: Data Structures and Algorithms (DSA), AI & ML Foundations, Computer Architecture, SQL, Web Development.

Patent App No: 202511131860 An IoT-based automatic water quality monitoring and dosing system.

CGPA: 8.13 | **Sem:** 5

ACHIEVEMENTS & CERTIFICATIONS

- Problem Solving:** Solved **200+ DSA problems** on LeetCode.
- Technical Workshop:** Attended workshop at **IIT Kanpur** on emerging technologies and applications.
- Certifications:** Infosys Springboard HTML, CSS, JavaScript, Python | Learning **Full Stack (Node.js, Express)**
- PROTECH 2K25:** Presented **Travel In-Depth** project at PSIT.